In [ ]:
```
import os
import nltk
nltk.download()
```

```
showing info https://raw.githubusercontent.com/nltk/nltk_data/gh-pages/index.xml
```

In [50]:
```
AI = '''Artificial Intelligence refers to the intelligence of machines. This is in
humans and animals. With Artificial Intelligence, machines perform functions such a
problem-solving. Most noteworthy, Artificial Intelligence is the simulation of huma
It is probably the fastest-growing development in the World of technology and innov
AI could solve major challenges and crisis situations.'''
```

In [51]:
```
AI
```

Out[51]:
```
'Artificial Intelligence refers to the intelligence of machines. This is in contra
st to the natural intelligence of \nhumans and animals. With Artificial Intelligen
ce, machines perform functions such as learning, planning, reasoning and \nproblem
-solving. Most noteworthy, Artificial Intelligence is the simulation of human inte
lligence by machines. \nIt is probably the fastest-growing development in the Worl
d of technology and innovation. Furthermore, many experts believe\nAI could solve
major challenges and crisis situations.'
```

In [53]:
```
type(AI)
```

Out[53]:
```
str
```

In [54]:
```
from nltk.tokenize import word_tokenize
```

In [55]:
```
AI_tokens = word_tokenize(AI)
AI_tokens
```

Out[55]:
```
['Artificial',
 'Intelligence',
 'refers',
 'to',
 'the',
 'intelligence',
 'of',
 'machines',
 '.',
 'This',
 'is',
 'in',
 'contrast',
 'to',
 'the',
 'natural',
 'intelligence',
 'of',
 'humans',
 'and',
 'animals',
 '.',
 'With',
 'Artificial',
 'Intelligence',
 ',',
 'machines',
 'perform',
 'functions',
 'such',
 'as',
 'learning',
 ',',
 'planning',
 ',',
 'reasoning',
 'and',
 'problem-solving',
 '.',
 'Most',
 'noteworthy',
 ',',
 'Artificial',
 'Intelligence',
 'is',
 'the',
 'simulation',
 'of',
 'human',
 'intelligence',
 'by',
 'machines',
 '.',
 'It',
 'is',
 'probably',
```

```
 'the',
 'fastest-growing',
 'development',
 'in',
 'the',
 'World',
 'of',
 'technology',
 'and',
 'innovation',
 '.',
 'Furthermore',
 ',',
 'many',
 'experts',
 'believe',
 'AI',
 'could',
 'solve',
 'major',
 'challenges',
 'and',
 'crisis',
 'situations',
 '.']
```

In [56]:
```
len(AI_tokens)
```

Out[56]:
```
81
```

In [57]:
```
from nltk.tokenize import sent_tokenize
```

In [58]:
```
AI_sent = sent_tokenize(AI)
AI_sent
```

Out[58]:
```
['Artificial Intelligence refers to the intelligence of machines.',
 'This is in contrast to the natural intelligence of \nhumans and animals.',
 'With Artificial Intelligence, machines perform functions such as learning, plann
ing, reasoning and \nproblem-solving.',
 'Most noteworthy, Artificial Intelligence is the simulation of human intelligence
by machines.',
 'It is probably the fastest-growing development in the World of technology and in
novation.',
 'Furthermore, many experts believe\nAI could solve major challenges and crisis si
tuations.']
```

In [59]:
```
len(AI_sent)
```

Out[59]:
```
6
```

In [60]:
```
AI
```

Out[60]:
```
'Artificial Intelligence refers to the intelligence of machines. This is in contra
st to the natural intelligence of \nhumans and animals. With Artificial Intelligen
ce, machines perform functions such as learning, planning, reasoning and \nproblem
-solving. Most noteworthy, Artificial Intelligence is the simulation of human inte
lligence by machines. \nIt is probably the fastest-growing development in the Worl
d of technology and innovation. Furthermore, many experts believe\nAI could solve
major challenges and crisis situations.'
```

In [61]:
```
from nltk.tokenize import blankline_tokenize # Give you hoe many Praragraph
AI_blank = blankline_tokenize(AI)
AI_blank
```

Out[61]:
```
['Artificial Intelligence refers to the intelligence of machines. This is in contr
ast to the natural intelligence of \nhumans and animals. With Artificial Intellige
nce, machines perform functions such as learning, planning, reasoning and \nproble
m-solving. Most noteworthy, Artificial Intelligence is the simulation of human int
elligence by machines. \nIt is probably the fastest-growing development in the Wor
ld of technology and innovation. Furthermore, many experts believe\nAI could solve
major challenges and crisis situations.']
```

In [62]:
```
len(AI_blank)
```

Out[62]:
```
1
```

In [ ]:
```

```

In [63]:
```
from nltk.tokenize import WhitespaceTokenizer
wt = WhitespaceTokenizer().tokenize(AI)
wt
```

Out[63]:
```
['Artificial',
 'Intelligence',
 'refers',
 'to',
 'the',
 'intelligence',
 'of',
 'machines.',
 'This',
 'is',
 'in',
 'contrast',
 'to',
 'the',
 'natural',
 'intelligence',
 'of',
 'humans',
 'and',
 'animals.',
 'With',
 'Artificial',
 'Intelligence,',
 'machines',
 'perform',
 'functions',
 'such',
 'as',
 'learning,',
 'planning,',
 'reasoning',
 'and',
 'problem-solving.',
 'Most',
 'noteworthy,',
 'Artificial',
 'Intelligence',
 'is',
 'the',
 'simulation',
 'of',
 'human',
 'intelligence',
 'by',
 'machines.',
 'It',
 'is',
 'probably',
 'the',
 'fastest-growing',
 'development',
 'in',
 'the',
 'World',
 'of',
 'technology',
```

```
 'and',
 'innovation.',
 'Furthermore,',
 'many',
 'experts',
 'believe',
 'AI',
 'could',
 'solve',
 'major',
 'challenges',
 'and',
 'crisis',
 'situations.']
```

In [64]:
```
print(len(wt))
```

```
70
```

In [65]:
```
len(AI_tokens)
```

Out[65]:
```
81
```

In [66]:
```
s = 'Good apple cost $3.88 in hyderbad. please buy two of them. thank.'
s
```

Out[66]:
```
'Good apple cost $3.88 in hyderbad. please buy two of them. thank.'
```

In [67]:
```
from nltk.tokenize import wordpunct_tokenize
wordpunct_tokenize(s)
```

Out[67]:
```
['Good',
 'apple',
 'cost',
 '$',
 '3',
 '.',
 '88',
 'in',
 'hyderbad',
 '.',
 'please',
 'buy',
 'two',
 'of',
 'them',
 '.',
 'thank',
 '.']
```

In [68]:
```
w_p = wordpunct_tokenize(AI)
w_p
```

Out[68]:
```
['Artificial',
 'Intelligence',
 'refers',
 'to',
 'the',
 'intelligence',
 'of',
 'machines',
 '.',
 'This',
 'is',
 'in',
 'contrast',
 'to',
 'the',
 'natural',
 'intelligence',
 'of',
 'humans',
 'and',
 'animals',
 '.',
 'With',
 'Artificial',
 'Intelligence',
 ',',
 'machines',
 'perform',
 'functions',
 'such',
 'as',
 'learning',
 ',',
 'planning',
 ',',
 'reasoning',
 'and',
 'problem',
 '-',
 'solving',
 '.',
 'Most',
 'noteworthy',
 ',',
 'Artificial',
 'Intelligence',
 'is',
 'the',
 'simulation',
 'of',
 'human',
 'intelligence',
 'by',
 'machines',
 '.',
 'It',
```

```
 'is',
 'probably',
 'the',
 'fastest',
 '-',
 'growing',
 'development',
 'in',
 'the',
 'World',
 'of',
 'technology',
 'and',
 'innovation',
 '.',
 'Furthermore',
 ',',
 'many',
 'experts',
 'believe',
 'AI',
 'could',
 'solve',
 'major',
 'challenges',
 'and',
 'crisis',
 'situations',
 '.']
```

In [69]:
```python
len(w_p)
```

Out[69]:
```
85
```

In [70]:
```python
import nltk
```

In [71]:
```python
from nltk.util import bigrams,trigrams,ngrams
```

In [77]:
```python
string = 'the best and most beautifull thing in the world cannot be seen or even to
quotes_tokens = nltk.word_tokenize(string)
quotes_tokens
```

Out[77]:
```
['the',
 'best',
 'and',
 'most',
 'beautifull',
 'thing',
 'in',
 'the',
 'world',
 'can',
 'not',
 'be',
 'seen',
 'or',
 'even',
 'touched',
 ',',
 'they',
 'must',
 'be',
 'felt',
 'with',
 'heart']
```

In [78]:
```
string
```

Out[78]:
```
'the best and most beautifull thing in the world cannot be seen or even touched,th
ey must be felt with heart'
```

In [74]:
```
quotes_tokens
```

Out[74]:
```
['weare', 'leareing', 'of', 'full', 'stack', 'data', 'science', 'from', 'nit']
```

In [79]:
```
len(quotes_tokens)
```

Out[79]:
```
23
```

In [80]:
```
quotes_bigrams = list(nltk.bigrams(quotes_tokens))
quotes_bigrams
```

Out[80]:
```
[('the', 'best'),
 ('best', 'and'),
 ('and', 'most'),
 ('most', 'beautifull'),
 ('beautifull', 'thing'),
 ('thing', 'in'),
 ('in', 'the'),
 ('the', 'world'),
 ('world', 'can'),
 ('can', 'not'),
 ('not', 'be'),
 ('be', 'seen'),
 ('seen', 'or'),
 ('or', 'even'),
 ('even', 'touched'),
 ('touched', ','),
 (',', 'they'),
 ('they', 'must'),
 ('must', 'be'),
 ('be', 'felt'),
 ('felt', 'with'),
 ('with', 'heart')]
```

In [81]:
```
quotes_tokens
```

Out[81]:
```
['the',
 'best',
 'and',
 'most',
 'beautifull',
 'thing',
 'in',
 'the',
 'world',
 'can',
 'not',
 'be',
 'seen',
 'or',
 'even',
 'touched',
 ',',
 'they',
 'must',
 'be',
 'felt',
 'with',
 'heart']
```

In [82]:
```
len(quotes_tokens)
```

Out[82]:
```
23
```

In [84]:
```
quotes_trigrams = list(nltk.trigrams(quotes_tokens))
quotes_trigrams
```

Out[84]:
```
[('the', 'best', 'and'),
 ('best', 'and', 'most'),
 ('and', 'most', 'beautifull'),
 ('most', 'beautifull', 'thing'),
 ('beautifull', 'thing', 'in'),
 ('thing', 'in', 'the'),
 ('in', 'the', 'world'),
 ('the', 'world', 'can'),
 ('world', 'can', 'not'),
 ('can', 'not', 'be'),
 ('not', 'be', 'seen'),
 ('be', 'seen', 'or'),
 ('seen', 'or', 'even'),
 ('or', 'even', 'touched'),
 ('even', 'touched', ','),
 ('touched', ',', 'they'),
 (',', 'they', 'must'),
 ('they', 'must', 'be'),
 ('must', 'be', 'felt'),
 ('be', 'felt', 'with'),
 ('felt', 'with', 'heart')]
```

In [85]:
```
quotes_ngrams = list(nltk.ngrams(quotes_tokens))
quotes_ngrams
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[85], line 1
----> 1 quotes_ngrams = list(nltk.ngrams(quotes_tokens))
      2 quotes_ngrams

TypeError: ngrams() missing 1 required positional argument: 'n'
```

In [86]:
```
quotes_ngrams = list(nltk.ngrams(quotes_tokens, 4))
quotes_ngrams
```

Out[86]:
```
[('the', 'best', 'and', 'most'),
 ('best', 'and', 'most', 'beautifull'),
 ('and', 'most', 'beautifull', 'thing'),
 ('most', 'beautifull', 'thing', 'in'),
 ('beautifull', 'thing', 'in', 'the'),
 ('thing', 'in', 'the', 'world'),
 ('in', 'the', 'world', 'can'),
 ('the', 'world', 'can', 'not'),
 ('world', 'can', 'not', 'be'),
 ('can', 'not', 'be', 'seen'),
 ('not', 'be', 'seen', 'or'),
 ('be', 'seen', 'or', 'even'),
 ('seen', 'or', 'even', 'touched'),
 ('or', 'even', 'touched', ','),
 ('even', 'touched', ',', 'they'),
 ('touched', ',', 'they', 'must'),
 (',', 'they', 'must', 'be'),
 ('they', 'must', 'be', 'felt'),
 ('must', 'be', 'felt', 'with'),
 ('be', 'felt', 'with', 'heart')]
```

In [89]:
```
quotes_ngrams = list(nltk.ngrams(quotes_tokens, 3))
quotes_ngrams
```

Out[89]:
```
[('the', 'best', 'and'),
 ('best', 'and', 'most'),
 ('and', 'most', 'beautifull'),
 ('most', 'beautifull', 'thing'),
 ('beautifull', 'thing', 'in'),
 ('thing', 'in', 'the'),
 ('in', 'the', 'world'),
 ('the', 'world', 'can'),
 ('world', 'can', 'not'),
 ('can', 'not', 'be'),
 ('not', 'be', 'seen'),
 ('be', 'seen', 'or'),
 ('seen', 'or', 'even'),
 ('or', 'even', 'touched'),
 ('even', 'touched', ','),
 ('touched', ',', 'they'),
 (',', 'they', 'must'),
 ('they', 'must', 'be'),
 ('must', 'be', 'felt'),
 ('be', 'felt', 'with'),
 ('felt', 'with', 'heart')]
```

In [90]:
```
quotes_ngrams = list(nltk.ngrams(quotes_tokens, 5))
quotes_ngrams
```

Out[90]:
```
[('the', 'best', 'and', 'most', 'beautifull'),
 ('best', 'and', 'most', 'beautifull', 'thing'),
 ('and', 'most', 'beautifull', 'thing', 'in'),
 ('most', 'beautifull', 'thing', 'in', 'the'),
 ('beautifull', 'thing', 'in', 'the', 'world'),
 ('thing', 'in', 'the', 'world', 'can'),
 ('in', 'the', 'world', 'can', 'not'),
 ('the', 'world', 'can', 'not', 'be'),
 ('world', 'can', 'not', 'be', 'seen'),
 ('can', 'not', 'be', 'seen', 'or'),
 ('not', 'be', 'seen', 'or', 'even'),
 ('be', 'seen', 'or', 'even', 'touched'),
 ('seen', 'or', 'even', 'touched', ','),
 ('or', 'even', 'touched', ',', 'they'),
 ('even', 'touched', ',', 'they', 'must'),
 ('touched', ',', 'they', 'must', 'be'),
 (',', 'they', 'must', 'be', 'felt'),
 ('they', 'must', 'be', 'felt', 'with'),
 ('must', 'be', 'felt', 'with', 'heart')]
```

In [92]:
```
quotes_ngrams_1= list(nltk.ngrams(quotes_tokens, 5))
quotes_ngrams_1
```

Out[92]:
```
[('the', 'best', 'and', 'most', 'beautifull'),
 ('best', 'and', 'most', 'beautifull', 'thing'),
 ('and', 'most', 'beautifull', 'thing', 'in'),
 ('most', 'beautifull', 'thing', 'in', 'the'),
 ('beautifull', 'thing', 'in', 'the', 'world'),
 ('thing', 'in', 'the', 'world', 'can'),
 ('in', 'the', 'world', 'can', 'not'),
 ('the', 'world', 'can', 'not', 'be'),
 ('world', 'can', 'not', 'be', 'seen'),
 ('can', 'not', 'be', 'seen', 'or'),
 ('not', 'be', 'seen', 'or', 'even'),
 ('be', 'seen', 'or', 'even', 'touched'),
 ('seen', 'or', 'even', 'touched', ','),
 ('or', 'even', 'touched', ',', 'they'),
 ('even', 'touched', ',', 'they', 'must'),
 ('touched', ',', 'they', 'must', 'be'),
 (',', 'they', 'must', 'be', 'felt'),
 ('they', 'must', 'be', 'felt', 'with'),
 ('must', 'be', 'felt', 'with', 'heart')]
```

In [93]:
```
len(quotes_tokens)
```

Out[93]:
```
23
```

In [95]:
```
quotes_ngrams = list(nltk.ngrams(quotes_tokens, 9))
quotes_ngrams
```

Out[95]:
```
[('the', 'best', 'and', 'most', 'beautifull', 'thing', 'in', 'the', 'world'),
 ('best', 'and', 'most', 'beautifull', 'thing', 'in', 'the', 'world', 'can'),
 ('and', 'most', 'beautifull', 'thing', 'in', 'the', 'world', 'can', 'not'),
 ('most', 'beautifull', 'thing', 'in', 'the', 'world', 'can', 'not', 'be'),
 ('beautifull', 'thing', 'in', 'the', 'world', 'can', 'not', 'be', 'seen'),
 ('thing', 'in', 'the', 'world', 'can', 'not', 'be', 'seen', 'or'),
 ('in', 'the', 'world', 'can', 'not', 'be', 'seen', 'or', 'even'),
 ('the', 'world', 'can', 'not', 'be', 'seen', 'or', 'even', 'touched'),
 ('world', 'can', 'not', 'be', 'seen', 'or', 'even', 'touched', ','),
 ('can', 'not', 'be', 'seen', 'or', 'even', 'touched', ',', 'they'),
 ('not', 'be', 'seen', 'or', 'even', 'touched', ',', 'they', 'must'),
 ('be', 'seen', 'or', 'even', 'touched', ',', 'they', 'must', 'be'),
 ('seen', 'or', 'even', 'touched', ',', 'they', 'must', 'be', 'felt'),
 ('or', 'even', 'touched', ',', 'they', 'must', 'be', 'felt', 'with'),
 ('even', 'touched', ',', 'they', 'must', 'be', 'felt', 'with', 'heart')]
```

In [99]:
```
# Next we need to make some changes in tokens and that is called as stemming, stemm
# also we will see some root form of the word & limitation of the word

#poter-stemmer
from nltk.stem import PorterStemmer
pst = PorterStemmer()
```

In [100…
```
pst.stem('having')
```

Out[100…
```
'have'
```

In [101…
```
pst.stem('affection')
```

Out[101… 'affect'

In [102…
```
pst.stem('playing')
```

Out[102… 'play'

In [103…
```
pst.stem('maximum')
```

Out[103… 'maximum'

In [104…
```
pst.stem('cooking')
```

Out[104… 'cook'

In [106…
```
pst.stem('give')
```

Out[106… 'give'

In [109…
```
word_to_stem=['give','giving','given','gave','Gangarapu Ajay Adams']
for words in word_to_stem:
    print(words+':'+pst.stem(words))
```

```
give:give
giving:give
given:given
gave:gave
Gangarapu Ajay Adams:gangarapu ajay adam
```

In [112…
```
word_to_stem=['give','giving','given','gave','loving','thinking','AJAY Adam']
for words in word_to_stem:
    print(words +':'+ pst.stem(words))
```

```
give:give
giving:give
given:given
gave:gave
loving:love
thinking:think
AJAY Adam:ajay adam
```

In [113…
```
#another stemmer known as Lencastemmer stemmer and lets see what the different we w
#stem the same thing using Lencastemmer

from nltk.stem import LancasterStemmer
lst = LancasterStemmer()
for words in word_to_stem:
    print(words + ':' + lst.stem(words))

# lancasterstemmer is more aggresive then the porterstemmer
```

```
give:giv
giving:giv
given:giv
gave:gav
loving:lov
thinking:think
AJAY Adam:ajay adam
```

In [114…

```
words_to_stem=['give','giving','given','gave','thinking', 'loving', 'final', 'final
# i am giving these different words to stem, using porter stemmer we get the output

for words in words_to_stem:
    print(words+ ':' +pst.stem(words))
```

```
give:give
giving:give
given:given
gave:gave
thinking:think
loving:love
final:final
finalized:final
finally:final
```

In [115…

```
#snowball stemmer
from nltk.stem import SnowballStemmer
sbst = SnowballStemmer('english')
for words in words_to_stem:
    print(words+ ':' +sbst.stem(words))
```

```
give:give
giving:give
given:given
gave:gave
thinking:think
loving:love
final:final
finalized:final
finally:final
```

In [121…

```
from nltk.stem import wordnet
from nltk.stem import WordNetLemmatizer
word_lem = WordNetLemmatizer()
```

In [118…

```
words_to_stem
```

Out[118…

```
['give',
 'giving',
 'given',
 'gave',
 'thinking',
 'loving',
 'final',
 'finalized',
 'finally']
```

In [123…
```
for words in word_to_stem:
    print(words+ ':' +word_lem.lemmatize(words))
```

```
give:give
giving:giving
given:given
gave:gave
loving:loving
thinking:thinking
AJAY Adam:AJAY Adam
```

In [124…
```
pst.stem('final')
```

Out[124… 'final'

In [126…
```
lst.stem('finally')
```

Out[126… 'fin'

In [127…
```
sbst.stem('finalized')
```

Out[127… 'final'

In [128…
```
lst.stem('final')
```

Out[128… 'fin'

In [129…
```
lst.stem('finalized')
```

Out[129… 'fin'

In [132…
```
from nltk.corpus import stopwords
```

In [133…
```
stopwords.words('english')
```

Out[133…

```
['a',
 'about',
 'above',
 'after',
 'again',
 'against',
 'ain',
 'all',
 'am',
 'an',
 'and',
 'any',
 'are',
 'aren',
 "aren't",
 'as',
 'at',
 'be',
 'because',
 'been',
 'before',
 'being',
 'below',
 'between',
 'both',
 'but',
 'by',
 'can',
 'couldn',
 "couldn't",
 'd',
 'did',
 'didn',
 "didn't",
 'do',
 'does',
 'doesn',
 "doesn't",
 'doing',
 'don',
 "don't",
 'down',
 'during',
 'each',
 'few',
 'for',
 'from',
 'further',
 'had',
 'hadn',
 "hadn't",
 'has',
 'hasn',
 "hasn't",
 'have',
 'haven',
```

```
 "haven't",
 'having',
 'he',
 "he'd",
 "he'll",
 'her',
 'here',
 'hers',
 'herself',
 "he's",
 'him',
 'himself',
 'his',
 'how',
 'i',
 "i'd",
 'if',
 "i'll",
 "i'm",
 'in',
 'into',
 'is',
 'isn',
 "isn't",
 'it',
 "it'd",
 "it'll",
 "it's",
 'its',
 'itself',
 "i've",
 'just',
 'll',
 'm',
 'ma',
 'me',
 'mightn',
 "mightn't",
 'more',
 'most',
 'mustn',
 "mustn't",
 'my',
 'myself',
 'needn',
 "needn't",
 'no',
 'nor',
 'not',
 'now',
 'o',
 'of',
 'off',
 'on',
 'once',
 'only',
```

```
 'or',
 'other',
 'our',
 'ours',
 'ourselves',
 'out',
 'over',
 'own',
 're',
 's',
 'same',
 'shan',
 "shan't",
 'she',
 "she'd",
 "she'll",
 "she's",
 'should',
 'shouldn',
 "shouldn't",
 "should've",
 'so',
 'some',
 'such',
 't',
 'than',
 'that',
 "that'll",
 'the',
 'their',
 'theirs',
 'them',
 'themselves',
 'then',
 'there',
 'these',
 'they',
 "they'd",
 "they'll",
 "they're",
 "they've",
 'this',
 'those',
 'through',
 'to',
 'too',
 'under',
 'until',
 'up',
 've',
 'very',
 'was',
 'wasn',
 "wasn't",
 'we',
 "we'd",
```

```
 "we'll",
 "we're",
 'were',
 'weren',
 "weren't",
 "we've",
 'what',
 'when',
 'where',
 'which',
 'while',
 'who',
 'whom',
 'why',
 'will',
 'with',
 'won',
 "won't",
 'wouldn',
 "wouldn't",
 'y',
 'you',
 "you'd",
 "you'll",
 'your',
 "you're",
 'yours',
 'yourself',
 'yourselves',
 "you've"]
```

In [135…

```
len(stopwords.words('english'))
```

Out[135…

```
198
```

In [134…

```
stopwords.words('german')
```

Out[134…

```
['aber',
 'alle',
 'allem',
 'allen',
 'aller',
 'alles',
 'als',
 'also',
 'am',
 'an',
 'ander',
 'andere',
 'anderem',
 'anderen',
 'anderer',
 'anderes',
 'anderm',
 'andern',
 'anderr',
 'anders',
 'auch',
 'auf',
 'aus',
 'bei',
 'bin',
 'bis',
 'bist',
 'da',
 'damit',
 'dann',
 'der',
 'den',
 'des',
 'dem',
 'die',
 'das',
 'dass',
 'daß',
 'derselbe',
 'derselben',
 'denselben',
 'desselben',
 'demselben',
 'dieselbe',
 'dieselben',
 'dasselbe',
 'dazu',
 'dein',
 'deine',
 'deinem',
 'deinen',
 'deiner',
 'deines',
 'denn',
 'derer',
 'dessen',
```

```
             'dich',
             'dir',
             'du',
             'dies',
             'diese',
             'diesem',
             'diesen',
             'dieser',
             'dieses',
             'doch',
             'dort',
             'durch',
             'ein',
             'eine',
             'einem',
             'einen',
             'einer',
             'eines',
             'einig',
             'einige',
             'einigem',
             'einigen',
             'einiger',
             'einiges',
             'einmal',
             'er',
             'ihn',
             'ihm',
             'es',
             'etwas',
             'euer',
             'eure',
             'eurem',
             'euren',
             'eurer',
             'eures',
             'für',
             'gegen',
             'gewesen',
             'hab',
             'habe',
             'haben',
             'hat',
             'hatte',
             'hatten',
             'hier',
             'hin',
             'hinter',
             'ich',
             'mich',
             'mir',
             'ihr',
             'ihre',
             'ihrem',
             'ihren',
             'ihrer',
```

```
                  'ihres',
                  'euch',
                  'im',
                  'in',
                  'indem',
                  'ins',
                  'ist',
                  'jede',
                  'jedem',
                  'jeden',
                  'jeder',
                  'jedes',
                  'jene',
                  'jenem',
                  'jenen',
                  'jener',
                  'jenes',
                  'jetzt',
                  'kann',
                  'kein',
                  'keine',
                  'keinem',
                  'keinen',
                  'keiner',
                  'keines',
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                  'könnte',
                  'machen',
                  'man',
                  'manche',
                  'manchem',
                  'manchen',
                  'mancher',
                  'manches',
                  'mein',
                  'meine',
                  'meinem',
                  'meinen',
                  'meiner',
                  'meines',
                  'mit',
                  'muss',
                  'musste',
                  'nach',
                  'nicht',
                  'nichts',
                  'noch',
                  'nun',
                  'nur',
                  'ob',
                  'oder',
                  'ohne',
                  'sehr',
                  'sein',
                  'seine',
                  'seinem',
```

```
                 'seinen',
                 'seiner',
                 'seines',
                 'selbst',
                 'sich',
                 'sie',
                 'ihnen',
                 'sind',
                 'so',
                 'solche',
                 'solchem',
                 'solchen',
                 'solcher',
                 'solches',
                 'soll',
                 'sollte',
                 'sondern',
                 'sonst',
                 'über',
                 'um',
                 'und',
                 'uns',
                 'unsere',
                 'unserem',
                 'unseren',
                 'unser',
                 'unseres',
                 'unter',
                 'viel',
                 'vom',
                 'von',
                 'vor',
                 'während',
                 'war',
                 'waren',
                 'warst',
                 'was',
                 'weg',
                 'weil',
                 'weiter',
                 'welche',
                 'welchem',
                 'welchen',
                 'welcher',
                 'welches',
                 'wenn',
                 'werde',
                 'werden',
                 'wie',
                 'wieder',
                 'will',
                 'wir',
                 'wird',
                 'wirst',
                 'wo',
                 'wollen',
```

```
 'wollte',
 'würde',
 'würden',
 'zu',
 'zum',
 'zur',
 'zwar',
 'zwischen']
```

In [137…
```
len(stopwords.words('german'))
```

Out[137…
```
232
```

In [139…
```
stopwords.words('spanish')
```

Out[139…

```
['de',
 'la',
 'que',
 'el',
 'en',
 'y',
 'a',
 'los',
 'del',
 'se',
 'las',
 'por',
 'un',
 'para',
 'con',
 'no',
 'una',
 'su',
 'al',
 'lo',
 'como',
 'más',
 'pero',
 'sus',
 'le',
 'ya',
 'o',
 'este',
 'sí',
 'porque',
 'esta',
 'entre',
 'cuando',
 'muy',
 'sin',
 'sobre',
 'también',
 'me',
 'hasta',
 'hay',
 'donde',
 'quien',
 'desde',
 'todo',
 'nos',
 'durante',
 'todos',
 'uno',
 'les',
 'ni',
 'contra',
 'otros',
 'ese',
 'eso',
 'ante',
 'ellos',
```

```
                    'e',
                    'esto',
                    'mí',
                    'antes',
                    'algunos',
                    'qué',
                    'unos',
                    'yo',
                    'otro',
                    'otras',
                    'otra',
                    'él',
                    'tanto',
                    'esa',
                    'estos',
                    'mucho',
                    'quienes',
                    'nada',
                    'muchos',
                    'cual',
                    'poco',
                    'ella',
                    'estar',
                    'estas',
                    'algunas',
                    'algo',
                    'nosotros',
                    'mi',
                    'mis',
                    'tú',
                    'te',
                    'ti',
                    'tu',
                    'tus',
                    'ellas',
                    'nosotras',
                    'vosotros',
                    'vosotras',
                    'os',
                    'mío',
                    'mía',
                    'míos',
                    'mías',
                    'tuyo',
                    'tuya',
                    'tuyos',
                    'tuyas',
                    'suyo',
                    'suya',
                    'suyos',
                    'suyas',
                    'nuestro',
                    'nuestra',
                    'nuestros',
                    'nuestras',
                    'vuestro',
```

```
'vuestra',
'vuestros',
'vuestras',
'esos',
'esas',
'estoy',
'estás',
'está',
'estamos',
'estáis',
'están',
'esté',
'estés',
'estemos',
'estéis',
'estén',
'estaré',
'estarás',
'estará',
'estaremos',
'estaréis',
'estarán',
'estaría',
'estarías',
'estaríamos',
'estaríais',
'estarían',
'estaba',
'estabas',
'estábamos',
'estabais',
'estaban',
'estuve',
'estuviste',
'estuvo',
'estuvimos',
'estuvisteis',
'estuvieron',
'estuviera',
'estuvieras',
'estuviéramos',
'estuvierais',
'estuvieran',
'estuviese',
'estuvieses',
'estuviésemos',
'estuvieseis',
'estuviesen',
'estando',
'estado',
'estada',
'estados',
'estadas',
'estad',
'he',
'has',
```

```
'ha',
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'han',
'haya',
'hayas',
'hayamos',
'hayáis',
'hayan',
'habré',
'habrás',
'habrá',
'habremos',
'habréis',
'habrán',
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'habríamos',
'habríais',
'habrían',
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'habíamos',
'habíais',
'habían',
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'hubiste',
'hubo',
'hubimos',
'hubisteis',
'hubieron',
'hubiera',
'hubieras',
'hubiéramos',
'hubierais',
'hubieran',
'hubiese',
'hubieses',
'hubiésemos',
'hubieseis',
'hubiesen',
'habiendo',
'habido',
'habida',
'habidos',
'habidas',
'soy',
'eres',
'es',
'somos',
'sois',
'son',
'sea',
'seas',
'seamos',
'seáis',
```

```
               'sean',
               'seré',
               'serás',
               'será',
               'seremos',
               'seréis',
               'serán',
               'sería',
               'serías',
               'seríamos',
               'seríais',
               'serían',
               'era',
               'eras',
               'éramos',
               'erais',
               'eran',
               'fui',
               'fuiste',
               'fue',
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               'fueron',
               'fuera',
               'fueras',
               'fuéramos',
               'fuerais',
               'fueran',
               'fuese',
               'fueses',
               'fuésemos',
               'fueseis',
               'fuesen',
               'sintiendo',
               'sentido',
               'sentida',
               'sentidos',
               'sentidas',
               'siente',
               'sentid',
               'tengo',
               'tienes',
               'tiene',
               'tenemos',
               'tenéis',
               'tienen',
               'tenga',
               'tengas',
               'tengamos',
               'tengáis',
               'tengan',
               'tendré',
               'tendrás',
               'tendrá',
               'tendremos',
               'tendréis',
```

```
 'tendrán',
 'tendría',
 'tendrías',
 'tendríamos',
 'tendríais',
 'tendrían',
 'tenía',
 'tenías',
 'teníamos',
 'teníais',
 'tenían',
 'tuve',
 'tuviste',
 'tuvo',
 'tuvimos',
 'tuvisteis',
 'tuvieron',
 'tuviera',
 'tuvieras',
 'tuviéramos',
 'tuvierais',
 'tuvieran',
 'tuviese',
 'tuvieses',
 'tuviésemos',
 'tuvieseis',
 'tuviesen',
 'teniendo',
 'tenido',
 'tenida',
 'tenidos',
 'tenidas',
 'tened']
```

In [140…

```
len(stopwords.words('spanish'))
```

Out[140…

```
313
```

In [143…

```
stopwords.words('chinese')
```

Out[143…

```
['一',
 '一下',
 '一些',
 '一切',
 '一则',
 '一天',
 '一定',
 '一方面',
 '一旦',
 '一时',
 '一来',
 '一样',
 '一次',
 '一片',
 '一直',
 '一致',
 '一般',
 '一起',
 '一边',
 '一面',
 '万一',
 '上下',
 '上升',
 '上去',
 '上来',
 '上述',
 '上面',
 '下列',
 '下去',
 '下来',
 '下面',
 '不一',
 '不久',
 '不仅',
 '不会',
 '不但',
 '不光',
 '不单',
 '不变',
 '不只',
 '不可',
 '不同',
 '不够',
 '不如',
 '不得',
 '不怕',
 '不惟',
 '不成',
 '不拘',
 '不敢',
 '不断',
 '不是',
 '不比',
 '不然',
 '不特',
 '不独',
```

```
'不管',
'不能',
'不要',
'不论',
'不足',
'不过',
'不问',
'与',
'与其',
'与否',
'与此同时',
'专门',
'且',
'两者',
'严格',
'严重',
'个',
'个人',
'个别',
'中小',
'中间',
'丰富',
'临',
'为',
'为主',
'为了',
'为什么',
'为什麽',
'为何',
'为着',
'主张',
'主要',
'举行',
'乃',
'乃至',
'么',
'之',
'之一',
'之前',
'之后',
'之後',
'之所以',
'之类',
'乌乎',
'乎',
'乘',
'也',
'也好',
'也是',
'也罢',
'了',
'了解',
'争取',
'于',
'于是',
'于是乎',
```

```
'云云',
'互相',
'产生',
'人们',
'人家',
'什么',
'什么样',
'什麼',
'今后',
'今天',
'今年',
'今後',
'仍然',
'从',
'从事',
'从而',
'他',
'他人',
'他们',
'他的',
'代替',
'以',
'以上',
'以下',
'以为',
'以便',
'以免',
'以前',
'以及',
'以后',
'以外',
'以後',
'以来',
'以至',
'以至于',
'以致',
'们',
'任',
'任何',
'任凭',
'任务',
'企图',
'伟大',
'似乎',
'似的',
'但',
'但是',
'何',
'何况',
'何处',
'何时',
'作为',
'你',
'你们',
'你的',
'使得',
```

```
 '使用',
 '例如',
 '依',
 '依照',
 '依靠',
 '促进',
 '保持',
 '俺',
 '俺们',
 '倘',
 '倘使',
 '倘或',
 '倘然',
 '倘若',
 '假使',
 '假如',
 '假若',
 '做到',
 '像',
 '允许',
 '充分',
 '先后',
 '先後',
 '先生',
 '全部',
 '全面',
 '兮',
 '共同',
 '关于',
 '其',
 '其一',
 '其中',
 '其二',
 '其他',
 '其余',
 '其它',
 '其实',
 '其次',
 '具体',
 '具体地说',
 '具体说来',
 '具有',
 '再者',
 '再说',
 '冒',
 '冲',
 '决定',
 '况且',
 '准备',
 '几',
 '几乎',
 '几时',
 '凭',
 '凭借',
 '出去',
 '出来',
```

```
'出现',
'分别',
'则',
'别',
'别的',
'别说',
'到',
'前后',
'前者',
'前进',
'前面',
'加之',
'加以',
'加入',
'加强',
'十分',
'即',
'即令',
'即使',
'即便',
'即或',
'即若',
'却不',
'原来',
'又',
'及',
'及其',
'及时',
'及至',
'双方',
'反之',
'反应',
'反映',
'反过来',
'反过来说',
'取得',
'受到',
'变成',
'另',
'另一方面',
'另外',
'只是',
'只有',
'只要',
'只限',
'叫',
'叫做',
'召开',
'叮咚',
'可',
'可以',
'可是',
'可能',
'可见',
'各',
'各个',
```

```
'各人',
'各位',
'各地',
'各种',
'各级',
'各自',
'合理',
'同',
'同一',
'同时',
'同样',
'后来',
'后面',
'向',
'向着',
'吓',
'吗',
'否则',
'吧',
'吧哒',
'吱',
'呀',
'呃',
'呕',
'呗',
'呜',
'呜呼',
'呢',
'周围',
'呵',
'呸',
'呼哧',
'咋',
'和',
'咚',
'咦',
'咱',
'咱们',
'咳',
'哇',
'哈',
'哈哈',
'哉',
'哎',
'哎呀',
'哎哟',
'哗',
'哟',
'哦',
'哩',
'哪',
'哪个',
'哪些',
'哪儿',
'哪天',
'哪年',
```

```
'哪怕',
'哪样',
'哪边',
'哪里',
'哼',
'哼唷',
'唉',
'啊',
'啐',
'啥',
'啦',
'啪达',
'喂',
'喏',
'喔唷',
'嗡嗡',
'嗬',
'嗯',
'嗳',
'嘎',
'嘎登',
'嘘',
'嘛',
'嘻',
'嘿',
'因',
'因为',
'因此',
'因而',
'固然',
'在',
'在下',
'地',
'坚决',
'坚持',
'基本',
'处理',
'复杂',
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'多少',
'多数',
'多次',
'大力',
'大多数',
'大大',
'大家',
'大批',
'大约',
'大量',
'失去',
'她',
'她们',
'她的',
'好的',
'好象',
'如',
```

```
'如上所述',
'如下',
'如何',
'如其',
'如果',
'如此',
'如若',
'存在',
'宁',
'宁可',
'宁愿',
'宁肯',
'它',
'它们',
'它们的',
'它的',
'安全',
'完全',
'完成',
'实现',
'实际',
'宣布',
'容易',
'密切',
'对',
'对于',
'对应',
'将',
'少数',
'尔后',
'尚且',
'尤其',
'就',
'就是',
'就是说',
'尽',
'尽管',
'属于',
'岂但',
'左右',
'巨大',
'巩固',
'己',
'已经',
'帮助',
'常常',
'并',
'并不',
'并不是',
'并且',
'并没有',
'广大',
'广泛',
'应当',
'应用',
'应该',
```

```
'开外',
'开始',
'开展',
'引起',
'强烈',
'强调',
'归',
'当',
'当前',
'当时',
'当然',
'当着',
'形成',
'彻底',
'彼',
'彼此',
'往',
'往往',
'待',
'後来',
'後面',
'得',
'得出',
'得到',
'心里',
'必然',
'必要',
'必须',
'怎',
'怎么',
'怎么办',
'怎么样',
'怎样',
'怎麽',
'总之',
'总是',
'总的来看',
'总的来说',
'总的说来',
'总结',
'总而言之',
'恰恰相反',
'您',
'意思',
'愿意',
'慢说',
'成为',
'我',
'我们',
'我的',
'或',
'或是',
'或者',
'战斗',
'所',
'所以',
```

```
'所有',
'所谓',
'打',
'扩大',
'把',
'抑或',
'拿',
'按',
'按照',
'换句话说',
'换言之',
'据',
'掌握',
'接着',
'接著',
'故',
'故此',
'整个',
'方便',
'方面',
'旁人',
'无宁',
'无法',
'无论',
'既',
'既是',
'既然',
'时候',
'明显',
'明确',
'是',
'是否',
'是的',
'显然',
'显著',
'普通',
'普遍',
'更加',
'曾经',
'替',
'最后',
'最大',
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'最高',
'有',
'有些',
'有关',
'有利',
'有力',
'有所',
'有效',
'有时',
'有点',
'有的',
```

```
'有着',
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'望',
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'本着',
'来',
'来着',
'极了',
'构成',
'果然',
'果真',
'某',
'某个',
'某些',
'根据',
'根本',
'欢迎',
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'正如',
'正常',
'此',
'此外',
'此时',
'此间',
'毋宁',
'每',
'每个',
'每天',
'每年',
'每当',
'比',
'比如',
'比方',
'比较',
'毫不',
'没有',
'沿',
'沿着',
'注意',
'深入',
'清楚',
'满足',
'漫说',
'焉',
'然则',
'然后',
'然後',
'然而',
'照',
'照着',
'特别是',
'特殊',
'特点',
'现代',
```

```
'现在',
'甚么',
'甚而',
'甚至',
'用',
'由',
'由于',
'由此可见',
'的',
'的话',
'目前',
'直到',
'直接',
'相似',
'相信',
'相反',
'相同',
'相对',
'相对而言',
'相应',
'相当',
'相等',
'省得',
'看出',
'看到',
'看来',
'看看',
'看见',
'真是',
'真正',
'着',
'着呢',
'矣',
'知道',
'确定',
'离',
'积极',
'移动',
'突出',
'突然',
'立即',
'第',
'等',
'等等',
'管',
'紧接着',
'纵',
'纵令',
'纵使',
'纵然',
'练习',
'组成',
'经',
'经常',
'经过',
'结合',
```

```
'结果',
'给',
'绝对',
'继续',
'继而',
'维持',
'综上所述',
'罢了',
'考虑',
'者',
'而',
'而且',
'而况',
'而外',
'而已',
'而是',
'而言',
'联系',
'能',
'能否',
'能够',
'腾',
'自',
'自个儿',
'自从',
'自各儿',
'自家',
'自己',
'自身',
'至',
'至于',
'良好',
'若',
'若是',
'若非',
'范围',
'莫若',
'获得',
'虽',
'虽则',
'虽然',
'虽说',
'行为',
'行动',
'表明',
'表示',
'被',
'要',
'要不',
'要不是',
'要不然',
'要么',
'要是',
'要求',
'规定',
'觉得',
```

```
'认为',
'认真',
'认识',
'让',
'许多',
'论',
'设使',
'设若',
'该',
'说明',
'诸位',
'谁',
'谁知',
'赶',
'起',
'起来',
'起见',
'趁',
'趁着',
'越是',
'跟',
'转动',
'转变',
'转贴',
'较',
'较之',
'边',
'达到',
'迅速',
'过',
'过去',
'过来',
'运用',
'还是',
'还有',
'这',
'这个',
'这么',
'这么些',
'这么样',
'这么点儿',
'这些',
'这会儿',
'这儿',
'这就是说',
'这时',
'这样',
'这点',
'这种',
'这边',
'这里',
'这麽',
'进入',
'进步',
'进而',
'进行',
```

```
'连',
'连同',
'适应',
'适当',
'适用',
'逐步',
'逐渐',
'通常',
'通过',
'造成',
'遇到',
'遭到',
'避免',
'那',
'那个',
'那么',
'那么些',
'那么样',
'那些',
'那会儿',
'那儿',
'那时',
'那样',
'那边',
'那里',
'那麽',
'部分',
'鄙人',
'采取',
'里面',
'重大',
'重新',
'重要',
'鉴于',
'问题',
'防止',
'阿',
'附近',
'限制',
'除',
'除了',
'除此之外',
'除非',
'随',
'随着',
'随著',
'集中',
'需要',
'非但',
'非常',
'非徒',
'靠',
'顺',
'顺着',
'首先',
```

```
 '高兴',
 '是不是']
```

In [144…

```
len(stopwords.words('chinese'))
```

Out[144…

```
841
```

In [145…

```
stopwords.words('spanish')
```

Out[145…

```
['de',
 'la',
 'que',
 'el',
 'en',
 'y',
 'a',
 'los',
 'del',
 'se',
 'las',
 'por',
 'un',
 'para',
 'con',
 'no',
 'una',
 'su',
 'al',
 'lo',
 'como',
 'más',
 'pero',
 'sus',
 'le',
 'ya',
 'o',
 'este',
 'sí',
 'porque',
 'esta',
 'entre',
 'cuando',
 'muy',
 'sin',
 'sobre',
 'también',
 'me',
 'hasta',
 'hay',
 'donde',
 'quien',
 'desde',
 'todo',
 'nos',
 'durante',
 'todos',
 'uno',
 'les',
 'ni',
 'contra',
 'otros',
 'ese',
 'eso',
 'ante',
 'ellos',
```

```
 'e',
 'esto',
 'mí',
 'antes',
 'algunos',
 'qué',
 'unos',
 'yo',
 'otro',
 'otras',
 'otra',
 'él',
 'tanto',
 'esa',
 'estos',
 'mucho',
 'quienes',
 'nada',
 'muchos',
 'cual',
 'poco',
 'ella',
 'estar',
 'estas',
 'algunas',
 'algo',
 'nosotros',
 'mi',
 'mis',
 'tú',
 'te',
 'ti',
 'tu',
 'tus',
 'ellas',
 'nosotras',
 'vosotros',
 'vosotras',
 'os',
 'mío',
 'mía',
 'míos',
 'mías',
 'tuyo',
 'tuya',
 'tuyos',
 'tuyas',
 'suyo',
 'suya',
 'suyos',
 'suyas',
 'nuestro',
 'nuestra',
 'nuestros',
 'nuestras',
 'vuestro',
```

```
                'vuestra',
                'vuestros',
                'vuestras',
                'esos',
                'esas',
                'estoy',
                'estás',
                'está',
                'estamos',
                'estáis',
                'están',
                'esté',
                'estés',
                'estemos',
                'estéis',
                'estén',
                'estaré',
                'estarás',
                'estará',
                'estaremos',
                'estaréis',
                'estarán',
                'estaría',
                'estarías',
                'estaríamos',
                'estaríais',
                'estarían',
                'estaba',
                'estabas',
                'estábamos',
                'estabais',
                'estaban',
                'estuve',
                'estuviste',
                'estuvo',
                'estuvimos',
                'estuvisteis',
                'estuvieron',
                'estuviera',
                'estuvieras',
                'estuviéramos',
                'estuvierais',
                'estuvieran',
                'estuviese',
                'estuvieses',
                'estuviésemos',
                'estuvieseis',
                'estuviesen',
                'estando',
                'estado',
                'estada',
                'estados',
                'estadas',
                'estad',
                'he',
                'has',
```

```
                    'ha',
                    'hemos',
                    'habéis',
                    'han',
                    'haya',
                    'hayas',
                    'hayamos',
                    'hayáis',
                    'hayan',
                    'habré',
                    'habrás',
                    'habrá',
                    'habremos',
                    'habréis',
                    'habrán',
                    'habría',
                    'habrías',
                    'habríamos',
                    'habríais',
                    'habrían',
                    'había',
                    'habías',
                    'habíamos',
                    'habíais',
                    'habían',
                    'hube',
                    'hubiste',
                    'hubo',
                    'hubimos',
                    'hubisteis',
                    'hubieron',
                    'hubiera',
                    'hubieras',
                    'hubiéramos',
                    'hubierais',
                    'hubieran',
                    'hubiese',
                    'hubieses',
                    'hubiésemos',
                    'hubieseis',
                    'hubiesen',
                    'habiendo',
                    'habido',
                    'habida',
                    'habidos',
                    'habidas',
                    'soy',
                    'eres',
                    'es',
                    'somos',
                    'sois',
                    'son',
                    'sea',
                    'seas',
                    'seamos',
                    'seáis',
```

```
                'sean',
                'seré',
                'serás',
                'será',
                'seremos',
                'seréis',
                'serán',
                'sería',
                'serías',
                'seríamos',
                'seríais',
                'serían',
                'era',
                'eras',
                'éramos',
                'erais',
                'eran',
                'fui',
                'fuiste',
                'fue',
                'fuimos',
                'fuisteis',
                'fueron',
                'fuera',
                'fueras',
                'fuéramos',
                'fuerais',
                'fueran',
                'fuese',
                'fueses',
                'fuésemos',
                'fueseis',
                'fuesen',
                'sintiendo',
                'sentido',
                'sentida',
                'sentidos',
                'sentidas',
                'siente',
                'sentid',
                'tengo',
                'tienes',
                'tiene',
                'tenemos',
                'tenéis',
                'tienen',
                'tenga',
                'tengas',
                'tengamos',
                'tengáis',
                'tengan',
                'tendré',
                'tendrás',
                'tendrá',
                'tendremos',
                'tendréis',
```

```
 'tendrán',
 'tendría',
 'tendrías',
 'tendríamos',
 'tendríais',
 'tendrían',
 'tenía',
 'tenías',
 'teníamos',
 'teníais',
 'tenían',
 'tuve',
 'tuviste',
 'tuvo',
 'tuvimos',
 'tuvisteis',
 'tuvieron',
 'tuviera',
 'tuvieras',
 'tuviéramos',
 'tuvierais',
 'tuvieran',
 'tuviese',
 'tuvieses',
 'tuviésemos',
 'tuvieseis',
 'tuviesen',
 'teniendo',
 'tenido',
 'tenida',
 'tenidos',
 'tenidas',
 'tened']
```

In [146…

```
len(stopwords.words('spanish'))
```

Out[146…

```
313
```

In [147…

```
stopwords.words('french')
```

Out[147…
```
['au',
 'aux',
 'avec',
 'ce',
 'ces',
 'dans',
 'de',
 'des',
 'du',
 'elle',
 'en',
 'et',
 'eux',
 'il',
 'ils',
 'je',
 'la',
 'le',
 'les',
 'leur',
 'lui',
 'ma',
 'mais',
 'me',
 'même',
 'mes',
 'moi',
 'mon',
 'ne',
 'nos',
 'notre',
 'nous',
 'on',
 'ou',
 'par',
 'pas',
 'pour',
 'qu',
 'que',
 'qui',
 'sa',
 'se',
 'ses',
 'son',
 'sur',
 'ta',
 'te',
 'tes',
 'toi',
 'ton',
 'tu',
 'un',
 'une',
 'vos',
 'votre',
 'vous',
```

```
                'c',
                'd',
                'j',
                'l',
                'à',
                'm',
                'n',
                's',
                't',
                'y',
                'été',
                'étée',
                'étées',
                'étés',
                'étant',
                'étante',
                'étants',
                'étantes',
                'suis',
                'es',
                'est',
                'sommes',
                'êtes',
                'sont',
                'serai',
                'seras',
                'sera',
                'serons',
                'serez',
                'seront',
                'serais',
                'serait',
                'serions',
                'seriez',
                'seraient',
                'étais',
                'était',
                'étions',
                'étiez',
                'étaient',
                'fus',
                'fut',
                'fûmes',
                'fûtes',
                'furent',
                'sois',
                'soit',
                'soyons',
                'soyez',
                'soient',
                'fusse',
                'fusses',
                'fût',
                'fussions',
                'fussiez',
                'fussent',
```

```
 'ayant',
 'ayante',
 'ayantes',
 'ayants',
 'eu',
 'eue',
 'eues',
 'eus',
 'ai',
 'as',
 'avons',
 'avez',
 'ont',
 'aurai',
 'auras',
 'aura',
 'aurons',
 'aurez',
 'auront',
 'aurais',
 'aurait',
 'aurions',
 'auriez',
 'auraient',
 'avais',
 'avait',
 'avions',
 'aviez',
 'avaient',
 'eut',
 'eûmes',
 'eûtes',
 'eurent',
 'aie',
 'aies',
 'ait',
 'ayons',
 'ayez',
 'aient',
 'eusse',
 'eusses',
 'eût',
 'eussions',
 'eussiez',
 'eussent']
```

In [148…

```
len(stopwords.words('french'))
```

Out[148…

```
157
```

In [149…

```
stopwords.words('Telugu')
```

```
---------------------------------------------------------------------------
OSError                                   Traceback (most recent call last)
Cell In[149], line 1
----> 1 stopwords.words(         )

File C:\ProgramData\anaconda3\Lib\site-packages\nltk\corpus\reader\wordlist.py:21, i
n WordListCorpusReader.words(self, fileids, ignore_lines_startswith)
     18 def words(self, fileids=None, ignore_lines_startswith="\n"):
     19     return [
     20         line
---> 21         for line in line_tokenize(self.raw(fileids))
     22         if not line.startswith(ignore_lines_startswith)
     23     ]

File C:\ProgramData\anaconda3\Lib\site-packages\nltk\corpus\reader\api.py:218, in Co
rpusReader.raw(self, fileids)
    216 contents = []
    217 for f in fileids:
--> 218     with self.open(f) as fp:
    219         contents.append(fp.read())
    220 return concat(contents)

File C:\ProgramData\anaconda3\Lib\site-packages\nltk\corpus\reader\api.py:231, in Co
rpusReader.open(self, file)
    223 """
    224 Return an open stream that can be used to read the given file.
    225 If the file's encoding is not None, then the stream will
   (...)    228 :param file: The file identifier of the file to read.
    229 """
    230 encoding = self.encoding(file)
--> 231 stream = self._root.join(file).open(encoding)
    232 return stream

File C:\ProgramData\anaconda3\Lib\site-packages\nltk\data.py:333, in FileSystemPathP
ointer.join(self, fileid)
    331 def join(self, fileid):
    332     _path = os.path.join(self._path, fileid)
--> 333     return FileSystemPathPointer(_path)

File C:\ProgramData\anaconda3\Lib\site-packages\nltk\data.py:311, in FileSystemPathP
ointer.__init__(self, _path)
    309 _path = os.path.abspath(_path)
    310 if not os.path.exists(_path):
--> 311     raise OSError("No such file or directory: %r" % _path)
    312 self._path = _path

OSError: No such file or directory: 'C:\\Users\\ANIL\\AppData\\Roaming\\nltk_data\\c
orpora\\stopwords\\Telugu'
```

In [150…

```
import re
punctuation =re.compile('[-.?!,:;()|0-9]')
```

In [152…

```
punctuation
```

Out[152…

```
re.compile(r'[-.?!,:;()|0-9]', re.UNICODE)
```

In [153…
```
AI
```

Out[153…
```
'Artificial Intelligence refers to the intelligence of machines. This is in contra
st to the natural intelligence of \nhumans and animals. With Artificial Intelligen
ce, machines perform functions such as learning, planning, reasoning and \nproblem
-solving. Most noteworthy, Artificial Intelligence is the simulation of human inte
lligence by machines. \nIt is probably the fastest-growing development in the Worl
d of technology and innovation. Furthermore, many experts believe\nAI could solve
major challenges and crisis situations.'
```

In [154…
```
AI_tokens
```

Out[154…
```
['Artificial',
 'Intelligence',
 'refers',
 'to',
 'the',
 'intelligence',
 'of',
 'machines',
 '.',
 'This',
 'is',
 'in',
 'contrast',
 'to',
 'the',
 'natural',
 'intelligence',
 'of',
 'humans',
 'and',
 'animals',
 '.',
 'With',
 'Artificial',
 'Intelligence',
 ',',
 'machines',
 'perform',
 'functions',
 'such',
 'as',
 'learning',
 ',',
 'planning',
 ',',
 'reasoning',
 'and',
 'problem-solving',
 '.',
 'Most',
 'noteworthy',
 ',',
 'Artificial',
 'Intelligence',
 'is',
 'the',
 'simulation',
 'of',
 'human',
 'intelligence',
 'by',
 'machines',
 '.',
 'It',
 'is',
 'probably',
```

```
 'the',
 'fastest-growing',
 'development',
 'in',
 'the',
 'World',
 'of',
 'technology',
 'and',
 'innovation',
 '.',
 'Furthermore',
 ',',
 'many',
 'experts',
 'believe',
 'AI',
 'could',
 'solve',
 'major',
 'challenges',
 'and',
 'crisis',
 'situations',
 '.']
```

In [155…

```
len(AI_tokens)
```

Out[155…

```
81
```

In [156…

```
# we will see how to work in POS using NLTK library
sent = 'Kathy is a natural when its comes to drawing'
sent_tokens = word_tokenize(sent)
sent_tokens
```

Out[156…

```
['Kathy', 'is', 'a', 'natural', 'when', 'its', 'comes', 'to', 'drawing']
```

In [157…

```
for token in sent_tokens:
    print(nltk.pos_tag([token]))
```

```
[('Kathy', 'NNP')]
[('is', 'VBZ')]
[('a', 'DT')]
[('natural', 'JJ')]
[('when', 'WRB')]
[('its', 'PRP$')]
[('comes', 'VBZ')]
[('to', 'TO')]
[('drawing', 'VBG')]
```

In [158…

```
sent2 = 'Ajay Adams Is Eating a delicious cake'
sent2_tokens = word_tokenize(sent2)

for token in sent2_tokens:
    print(nltk.pos_tag([token]))
```

```
[('Ajay', 'NN')]
[('Adams', 'NNS')]
[('Is', 'NN')]
[('Eating', 'VBG')]
[('a', 'DT')]
[('delicious', 'JJ')]
[('cake', 'NN')]
```

In [159...

```
from nltk import ne_chunk
```

In [162...

```
NE_sent = 'The US President Staya in The WHITEHOUSE'
```

In [164...

```
NE_tokens = word_tokenize(NE_sent)
NE_tokens
```

Out[164...

```
['The', 'US', 'President', 'Staya', 'in', 'The', 'WHITEHOUSE']
```

In [166...

```
NE_tages = nltk.pos_tag(NE_tokens)
NE_tages
```

Out[166...

```
[('The', 'DT'),
 ('US', 'NNP'),
 ('President', 'NNP'),
 ('Staya', 'NNP'),
 ('in', 'IN'),
 ('The', 'DT'),
 ('WHITEHOUSE', 'NNP')]
```

In [172...

```
NE_NER = ne_chunk(NE_tages)
print(NE_NER)
```

```
(S
  The/DT
  (ORGANIZATION US/NNP)
  President/NNP
  (PERSON Staya/NNP)
  in/IN
  The/DT
  (ORGANIZATION WHITEHOUSE/NNP))
```

In [173...

```
new = 'The big cat ate the little mouse who was after fresh cheese'
new_tokens = nltk.pos_tag(word_tokenize(new))
new_tokens
```

Out[173…

```
[('The', 'DT'),
 ('big', 'JJ'),
 ('cat', 'NN'),
 ('ate', 'VBD'),
 ('the', 'DT'),
 ('little', 'JJ'),
 ('mouse', 'NN'),
 ('who', 'WP'),
 ('was', 'VBD'),
 ('after', 'IN'),
 ('fresh', 'JJ'),
 ('cheese', 'NN')]
```

In [174…

```
# Libraries install -- pip install WordCloud
from wordcloud import WordCloud
import matplotlib.pyplot as plt
```

In [175…

```
# Create List Of Words
text=("Python Python python Matplotlib Seabron Network Plot Violin Chart Pandas Dat
```

In [176…

```
text
```

Out[176…

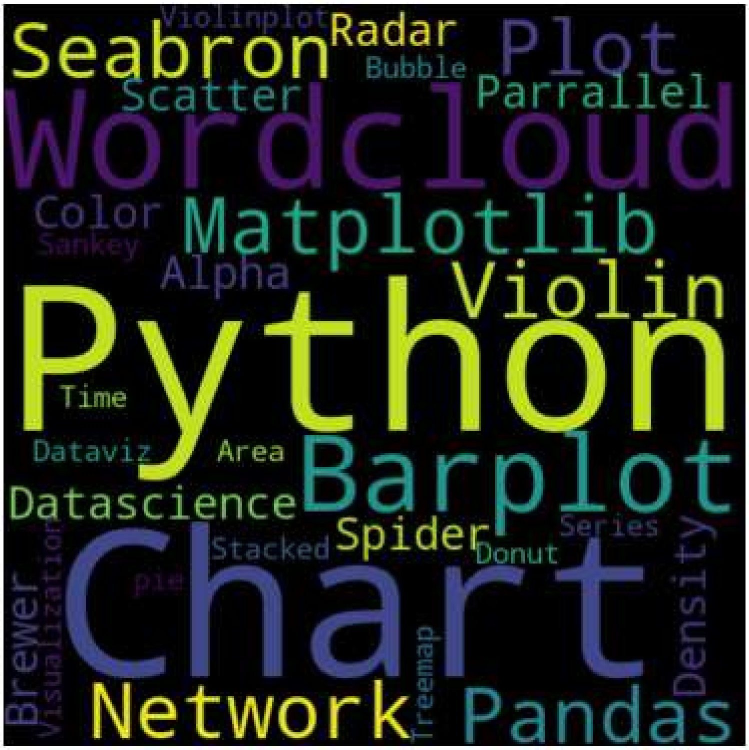
```
'Python Python python Matplotlib Seabron Network Plot Violin Chart Pandas Datascie
nce Wordcloud Spider Radar Parrallel Alpha Color Brewer Density Scatter Barplot Ba
rplot Violinplot Treemap Stacked Area Chart Chart Visualization Dataviz Donut pie
Time -Series Wordcloud Wordcloud Sankey Bubble'
```

In [177…

```
# Create THE WordCloud OBJECT
wordcloud = WordCloud(width=480, height=480, margin=0).generate(text)
```

In [180…

```
# Display The Generated Image
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.margins(x=0, y=0)
plt.show()
```


Violinplot
Radar
Seabron
Plot
Bubble
Scatter
Parrallel
Wordcloud
Color
Matplotlib
Sankey
Alpha
Violin
Python
Time
Dataviz
Area
Barplot
Datascience
Spider
Series
Stacked
Donut
pie
Density
Brewer
Visualization
Chart
Treemap
Network
Pandas

In [ ]: